Clinical Note Auditor

Paste a messy clinical note. Get a structured SOAP note, safety flags, and a checklist of what to fix before signing.

Caleb John Daniel Balakumar (solo)
DSH Hacks V2 · AI x Healthcare
github.com/calebjohndan-hoop/clinical-note-auditor

PROBLEM

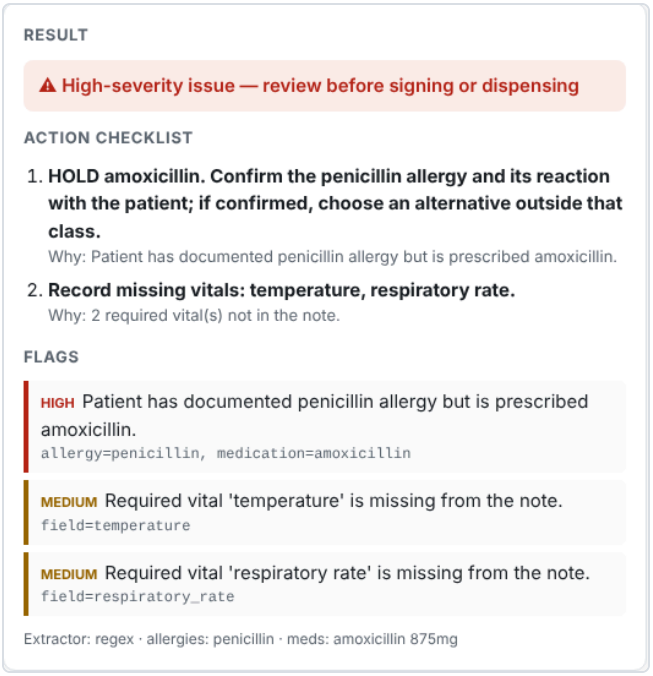
Clinical notes are written fast, in free text, full of abbreviations and brand names. A penicillin allergy on line 3 and an amoxicillin order on line 30 is exactly the error a tired reader misses. WHO estimates medication errors cost about **US$42 billion a year** worldwide. Most documentation tools check billing completeness, not medication safety inside the note.

WHAT IT DOES

- **Structured SOAP** version of the messy note.
- **Safety flags**: drug–allergy conflicts, out-of-range doses, missing vitals, missing or unrecognized allergy documentation. Every flag shows the evidence that triggered it.
- **Action checklist** in priority order, e.g. "HOLD amoxicillin. Confirm the penicillin allergy and its reaction with the patient."

HOW IT WORKS: THE LLM READS, THE RULES DECIDE

- An extractor turns free text into fields (allergies, medications and doses, vitals): **regex**, **LLM** (Groq gpt-oss-120b or Gemini), or **hybrid** (both).
- Every flag comes from **deterministic rules** over a reference table where each entry is sourced (FDA labels via DailyMed, AAAAI, ACR, AHA).
- **The LLM can only add findings, never remove them.** Hybrid takes the union of regex and LLM findings. A test runs the whole safety suite against an LLM that returns nothing, and it still passes.
- **Failures are loud.** If the LLM fails the screen says so; it never shows a silent all-clear.
- **Clinically informed choices:** penicillin allergy does not block cephalosporins and "iodine allergy" does not block contrast (both myths); codeine→hydrocodone is "review", not "stop".



Real output for "Pt has an allergy to PCN (hives)... Rx: Augmentin 875 mg". PCN and Augmentin are recognized as penicillin and amoxicillin.

EVIDENCE ON REAL NOTES

I hand-labeled 25 real de-identified MTSamples notes (allergies, medications and doses, vitals present) and re-checked every label on a second pass. 10 notes were held out until the end.

HELD-OUT: 10 UNSEEN NOTES	REGEX	HYBRID
Allergies (precision / recall)	88 / 100	88 / 100
Medications (precision / recall)	68 / 61	76 / **100**
Safety flags (precision / recall)	91 / 91	91 / 91

The LLM raised medication recall from 61% to 100% on unseen notes.

After fixing the regex bugs those notes exposed: medications 93 / 96 and allergies and flags 100 / 100 across all 25 (no longer held-out). 97 automated tests. I found and corrected one of my own label errors while re-checking.

LIMITATIONS

Small sample (25 notes, one labeler); flag scores are derived from my labels, so they mostly measure extraction. Dose table covers 7 drugs and allergy table 7 classes because every entry must be sourced; dose checks are per-dose sanity bounds, not personalized dosing. Misses drugs with no mg dose, counts supplements as medications, and can't tell a past or planned dose from a current one. **Research prototype on de-identified data. Not a medical device. Not clinically validated.**

NEXT

Multi-labeler evaluation on more notes; more sourced drugs; kidney-function dosing; a pre-sign check inside the EHR note editor.

Stack: Python, FastAPI, regex + LLM extraction, deterministic rule engine, pytest

Built with AI coding assistance; problem choice, labeling and evaluation are my own.